

SON LA, Viet Nam

WMO: 488060

Lat: 21.3322N

Lon: 103.9060E

Elev: 676

StdP: 93.46

Time Zone: 7.00 (E07)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.1	8.5	2.4	4.9	12.0	4.4	5.6	12.3	7.0	14.4	6.0	15.7	1.3	140	0.253

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.6	33.2	22.3	32.2	22.8	31.3	23.2	25.6	29.8	25.2	29.3	24.8	28.8	2.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.5	21.1	27.6	24.1	20.6	27.0	23.7	20.2	26.6	83.4	30.1	81.6	29.6	80.1	29.1	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.8	5.0	4.2	4.3	34.9	1.8	0.9	3.1	35.6	2.0	36.1	1.0	36.7	-0.3	37.3
			3.6	27.2	1.6	1.1	2.5	28.0	1.6	28.7	0.7	29.3	-0.5	30.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.3	16.0	18.2	21.4	24.3	25.5	26.1	25.9	25.7	24.9	23.0	19.8	16.5
	DBStd	4.48	3.38	4.16	3.48	2.86	2.24	1.49	1.30	1.22	1.52	1.93	2.71	3.01
	HDD10.0	4	2	2	0	0	0	0	0	0	0	0	0	1
	HDD18.3	239	86	49	16	1	0	0	0	0	0	1	16	71
	CDD10.0	4494	187	231	354	430	482	484	492	488	447	404	293	203
	CDD18.3	1687	12	45	112	181	223	234	233	229	197	146	59	15
	CDH23.3	12307	107	363	850	1519	1819	1878	1684	1609	1304	792	306	76
CDH26.7	3730	11	92	311	626	676	564	458	443	348	157	41	4	
Wind	WSAvg	1.5	1.6	1.9	1.9	1.7	1.5	1.4	1.3	1.2	1.1	1.2	1.3	1.5
Precipitation	PrecAvg	1396	26	24	53	106	198	214	273	257	129	55	35	27
	PrecMax	2026	103	76	135	278	380	377	491	418	421	171	143	113
	PrecMin	1037	0	0	0	40	48	85	135	109	41	0	0	1
	PrecStd	231	30	23	33	49	83	80	86	92	76	40	39	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.0	30.7	33.3	34.8	34.7	33.1	32.5	32.2	31.8	30.6	29.2	27.2
		MCWB	16.8	18.1	18.7	20.9	22.6	24.2	25.2	24.9	24.3	23.0	21.4	19.4
	2%	DB	26.1	29.3	31.8	33.2	33.1	31.6	31.2	31.1	30.8	29.4	27.9	25.3
		MCWB	17.4	17.8	19.1	21.8	23.2	24.5	24.8	24.7	24.0	22.5	20.7	18.4
	5%	DB	24.3	27.8	30.1	32.0	31.8	30.7	30.2	30.1	29.8	28.5	26.7	23.8
		MCWB	17.3	18.2	19.5	22.1	23.4	24.4	24.6	24.5	23.6	22.2	20.4	18.0
	10%	DB	22.2	25.7	28.3	30.5	30.4	29.7	29.2	29.2	28.9	27.4	25.2	22.0
		MCWB	16.7	18.2	19.9	22.2	23.3	24.2	24.5	24.3	23.2	21.8	19.4	17.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.9	21.4	22.7	24.8	25.4	26.0	26.1	25.9	25.4	24.4	23.0	20.8
		MCDB	25.2	27.4	29.0	31.0	30.6	30.5	30.3	30.0	29.8	28.6	27.2	24.3
	2%	WB	18.6	20.2	21.8	24.0	24.8	25.3	25.6	25.3	24.7	23.5	22.1	19.8
		MCDB	23.3	26.0	27.7	29.8	29.7	29.5	29.5	29.4	28.9	27.8	26.0	23.7
	5%	WB	17.8	19.4	21.1	23.3	24.3	24.9	25.1	24.9	24.3	22.9	21.2	18.6
		MCDB	22.3	24.7	26.8	29.1	29.1	28.9	28.9	28.9	28.2	27.0	24.8	22.5
	10%	WB	17.0	18.5	20.5	22.8	23.9	24.6	24.7	24.5	23.9	22.4	20.4	17.6
		MCDB	21.2	23.5	25.6	28.2	28.5	28.2	28.1	28.1	27.4	26.0	23.8	21.4
Mean Daily Temperature Range	5% DB	MDBR	8.8	10.0	10.1	9.4	8.1	6.6	6.2	6.5	7.5	7.9	9.0	9.0
		MCDBR	13.5	14.8	14.3	12.0	10.3	8.2	8.0	8.3	9.1	9.5	10.9	11.7
		MCWBR	5.7	5.2	4.7	3.7	3.0	2.6	2.9	3.0	3.2	3.4	4.5	5.3
	5% WB	MCDBR	11.0	12.2	11.4	10.5	8.4	7.1	7.0	7.2	7.7	8.1	9.5	10.1
		MCWBR	5.3	5.2	4.7	4.1	3.3	2.9	3.0	3.0	3.0	3.4	4.5	5.0
Clear-Sky Solar Irradiance	taub	0.475	0.538	0.761	0.741	0.488	0.447	0.444	0.487	0.517	0.563	0.485	0.498	
	taud	2.011	1.871	1.444	1.535	2.188	2.332	2.328	2.182	2.067	1.885	2.039	1.947	
	Ebn at Noon	783	758	615	640	819	846	849	817	788	726	768	741	
	Edh at Noon	161	197	314	290	150	128	129	150	166	191	156	167	
All-Sky Solar Radiation	RadAvg	2.96	3.82	4.20	4.88	4.89	4.38	4.39	4.46	4.38	3.73	3.60	2.96	
	RadStd	0.43	0.61	0.37	0.49	0.45	0.37	0.39	0.32	0.23	0.29	0.38	0.34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+0.57	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	+0.35	N/A	-0.78	+0.62	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page